

Experiment: Mechanical Properties of Elastomers

Part 1: Making the samples

Procedure for manufacturing of polydimethylsiloxane tensile test samples

1. Check the formulation that has been assigned to your group and the sample code (see Table below). All these formulations Sylgard 184 resin and a crosslinker. The resin and crosslinker formulations are viscous (they flow slowly) and are sticky.
2. Take a clean plastic cup and immediately label it with your group number and the corresponding sample code. Make sure the label on the cup can be read easily with group number **and** sample code and securely adhered to the cup. Make sure you replace it with a new label if this is not the case.
3. Check the target weight of the resin from the formulation table. Place the labelled cup onto the balance and zero the balance (this means to make sure the balance reader shows "0" before you weigh anything else) (balances are sometimes unstable, so the numbers may oscillate a little bit).
4. Slowly, pour in the resin into the cup, allowing time for the balance to equilibrate. Pour in even more slowly if the reading is close to the target weight. Try to get as close as possible to the target weight. Make sure that you record the weight that you actually measured out in the experimental Excel form (**this is NOT your target weight, it is the exact weight that you have just measured with the balance**). If you go over the target weight, do not discard your formulation, discuss what to do with the teachers. Input the actual weight into the Excel form.
5. *Make sure that you clean any surfaces contaminated with the resin (including the balance).* At this point, if you are not ready to continue weighing the crosslinker, allow another group to use the balance if necessary.
6. *Clean any contaminated surface.* At this point, if you are not ready to continue weighing the crosslinker, allow another group to use the balance if necessary.
7. Check the target weight of the crosslinker from the formulation table. Zero the balance again and weigh in the crosslinker, exactly as explained above for the resin(s). Make sure you record the **exact** weight of the crosslinker in the experimental Excel form and *clean any contaminated surfaces*.
8. Take a clean chopstick and use it to mix your formulation well. Try not to spill the formulation while you mix the cup content. After mixing well, make sure that you allow as much of the formulation still sticking to the chopstick to drop inside the cup.
9. Dispose the chopstick in the appropriate bin. Make sure ALL surfaces are clean.

10. Take your correctly labelled formulation cup to the degassing equipment. Your teacher will degass your sample together with some of the other groups. Observe what happens during this process.
11. Take a clean mould sheet and label it with your group number **and** sample code.
12. Take the cup with your group formulation and slowly pour in formulation in one of the grooves in the mould. Stop at intervals to allow the viscous formulation to settle (this will show you how much more formulation is required until the groove is full). Pour even more slowly as you get close to filling each groove up. Do not let the grooves overflow with formulation.
13. Repeat until you have filled in 4 grooves. If there is extra formulation left after you have finished pouring into the 4 grooves give it back to the teachers.
14. Allow the resin to settle. Avoid moving or shaking the mould. If you need to move it, do it GENTLY and STEADILY, making sure you do not spill any resin. Alert your teacher as you need to take your mould to one of the laboratory ovens to cure (“curing” the resin means to let it harden: what process is taking place in the resin matrix during curing?). Make sure the mould is correctly identified with your group number AND sample code. The teachers will look after your mould until the samples are fully set.
15. Make sure that one member of your group has correctly filled in all the relevant information in the group’s experimental form. The data belongs to the group, so every member of the group sure be sure that he data is correct. Make sure **everyone** in your group has access to the data by emailing the completed form to them. **Always use Queen Mary email accounts to email your data.**
16. Before you leave the laboratory: make sure you have cleaned any contaminated surfaces, disposed of gloves and contaminated material in the appropriate bin, and wash your hands.
17. Select **one** person in your group to upload the completed experimental form on QMPLUS, as indicated in your classes. Your Excel file (do not use any other file type) should be labelled as follows: “GroupXX SampleCode IntroEngMat Experiment1 Formulation data” (XX is your group number, including M or P for your programme, Sample code is the one on the formulation table, see Table 1).

Part 2: Testing the samples

Procedure for mechanical testing of polydimethylsiloxane tensile samples

When you enter the laboratory, find where your group will be working. You will have access to the following equipment, the samples that you made in the last lab; a metal frame to hang your samples; a balance; a ruler; some clips; and various things you can use as weights – see Figure 1.

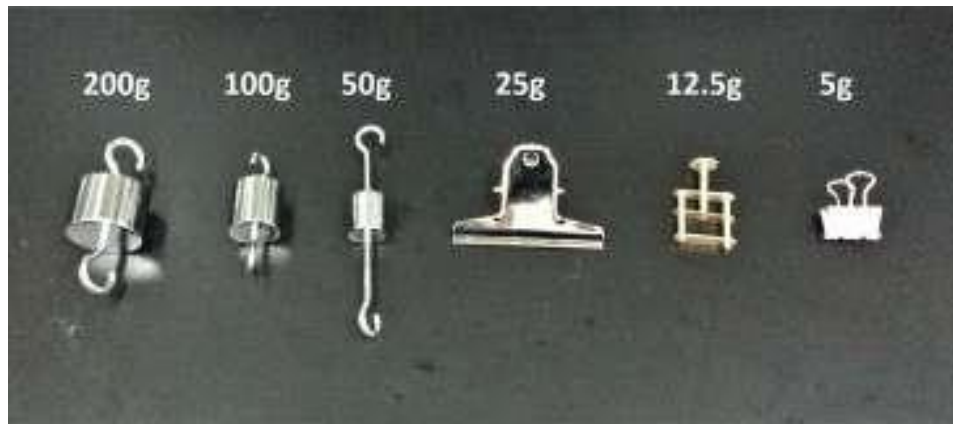


Figure 1: Various items that you can use to load your sample (Approximate weights shown: you need to measure them).

1. Check the formulation that has been assigned to your group and the sample code (see Table below) and make sure you are working with the right samples. If the sample is tacky (sticky) wear gloves.
2. Make sure your group information and sample code is entered correctly in the Experimental Form in cells B3 and B4, and that the exact mass of resin and crosslinker you use appear on cells B7 and B8.
3. Take your **first sample** from the mould and write the sample number on both ends of the sample using a pen. Enter the sample number into your Experimental form in cell E3. (This will be your sample A, so make sure this is added to the sample code).
4. Measure the **thickness** and the **width** of the sample over the straight part and immediately record it on the Experimental form in your computer on cells E5 and E6. Save the file.
5. **Draw two lines** across the width of the sample. These should be on the **straight part** of the sample, as far apart as possible but not within the curved section. Measure the distance between the two lines in mm using the ruler and immediately record it on the Experimental form in cell E10 (the weight recorded in cell D10 should be 0). Save the file.
6. **Clamp the sample** firmly at one end, using one of the paper clips or clamps provided, and hang it from the retort stand.

7. **Weigh a second clamp** using the balance and secure it from the bottom of the sample. Record the weight of the clamp immediately in cell D11 in the Experimental form. Save the file.
8. **Measure the length** between the two lines in mm using the ruler and record it immediately in the Experimental form in your computer in cell E11. Save the file.

Now **the group** needs to consider how much weight to add to the sample. It is important to think carefully about the expected relationship between elastomer extension and the formulation of your specific sample, in order to decide on what weight you will add first.

You need to take 10 measurements of load and extension for each sample: therefore, it is important you add small weight increments at a time, aiming at not more than 100 % extension at the 10th measurement.

It is alright if the first or second weight is not appropriate (too much extension or too little), as, for the following sample, you can check and adjust according to your discussions in your groups (however, do not take weights off a sample, once you have added them). If your first sample breaks by putting too much weight on it, do not worry, as you still have 3 more samples to work with! (Make sure you make suitable changes so that you do not break all samples!).

9. **Choose a first weight** according to the discussion in your group. Weigh the first weight, add it to the clamp weight, and record the accumulated weight (total weight on your sample) immediately on the Experimental form in cell D12 (and save the file).
10. Make sure a person in your group is ready to catch the sample or weight if it drops, then slowly hang the weight from the bottom clamp and slowly release it. Keep checking the sample or weights do not drop suddenly or the sample does not slide out of the lower clamp.
11. **Measure the extension** of the sample between the marked lines using the ruler and record it immediately in the Experimental form in cell E12 (and save the file).
12. Based on the amount of extension and your previous discussions, **choose a second weight** and weigh it on the balance. **Add this weight** to the previous accumulated weight (see cell D12 on Experimental form) and record immediately the new accumulated weight on cell D13. Save the file.
13. Slowly hang the weight from the previous weight or the bottom clamp and slowly let the sample take up the new weight.
14. **Measure the extension** of the sample between the marked lines using the ruler and record it immediately in the Experimental form in cell E13 (and save the file).

Repeat steps 11-13 adding successive extra weights, recording in the corresponding cells on the Experimental form, until you have 10 successive weight and length measurements, or the sample has failed. If you observe that the sample is slipping out of the clamp, consider removing the

weights and securing a firmer clamp (ask the teachers if this is the case). Do not remove the weights, otherwise.

If you decide to measure two samples at once (this is possible using the frames you will be provided) you should make sure that you still **make decisions as one team**. This means, that **everyone is following the same strategy** (what weights to add, and in what sequence, etc). Also, make sure you do not use too many weights as all groups need them.

Record any notes or observations about your sample in the Experimental form cell D23. For example: *'the sample broke at a bubble with a load of 53.4g'*; or, *'the sample slipped out of the clamp with a load of 205g and we changed to a larger clamp'*.

It is important that you **record what actually happened** in the laboratory. This is good practice in Science and Engineering. No result is a bad result. All results contribute to our learning and understanding.

15. Based on the previous results, discuss in your group any changes that you should consider to make sure a steady extension of not more than 100% after 10 measurements.

16. **Repeat steps 2-13** for your other 3 samples (entering the data on the experimental form in the corresponding cells, now in columns G, H, etc).

17. Return all the samples to the teachers in the labelled bag provided. **Tidy up your bench.**

Before you leave the laboratory, select **one** person in your group to **submit the completed Experimental form** to QMPLUS: Your Excel file (do not use any other file type) should be labelled as follows: "GroupXX SampleCode IntroEngMat Experiment1 Formulation data" (XX is your group number, including M or P for your programme, Sample code is the one on the formulation table, see Table 1).